

4.4 Surface Area; Parametric Surfaces

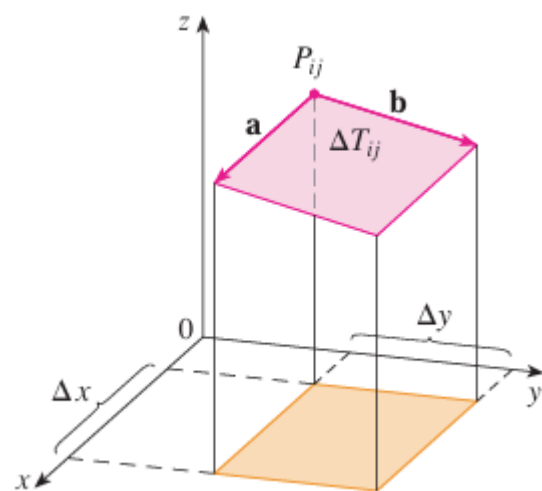
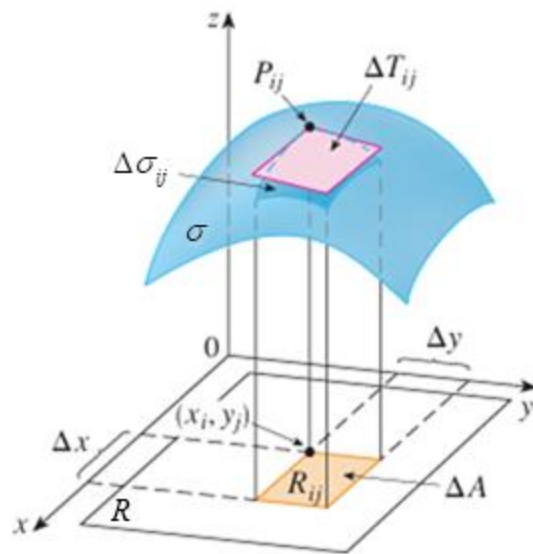
Motivation: Suppose that σ is a surface with equation $z = f(x, y)$ where f has continuous partial derivatives. Suppose also that $f(x, y) \geq 0$, and we wish to find the surface area of σ that exists over some region R in the xy -plane.

Our strategy for finding the surface area will be to divide R up into small rectangles R_{ij} with area $\Delta A = \Delta x \Delta y$ and thereby split up σ into smaller “patches” which we will call $\Delta \sigma_{ij}$. We then want to approximate the area of each $\Delta \sigma_{ij}$ over the rectangle R_{ij} . A picture of what a $\Delta \sigma_{ij}$ over R_{ij} might look like is shown at the right.

In actuality there would be many “pieces” $\Delta \sigma_{ij}$ of σ like the one pictured because there would be many rectangles R_{ij} . We are just focusing on one of the pieces.

Our plan is to now consider the point $P_{ij}(x_i, y_j)$ in R_{ij} closest to the origin, then $(x_i, y_j, f(x_i, y_j))$ is a point on the surface directly above the corner of R_{ij} closest to the origin. At this point we will construct a tangent plane to the surface (shown in pink above) and we will use the area of the tangent plane ΔT_{ij} (which is just the area of a parallelogram) to estimate the area of the surface $\Delta \sigma_{ij}$.

Now suppose we build vectors $\mathbf{a}$ and $\mathbf{b}$ that start at P_{ij} and lie along the sides of the parallelogram (as shown in the image at the right). We let ΔT_{ij} represent the area of this parallelogram. We note that as R_{ij} gets smaller and smaller ΔT_{ij} will do a better job of approximating the “piece” of σ . This leads to the following definition.

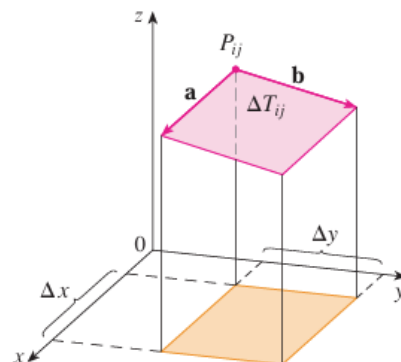


Definition: The surface area of σ (as described above) is
$$S = \lim_{m, n \rightarrow \infty} \sum_{i=1}^m \sum_{j=1}^n \Delta T_{ij}.$$

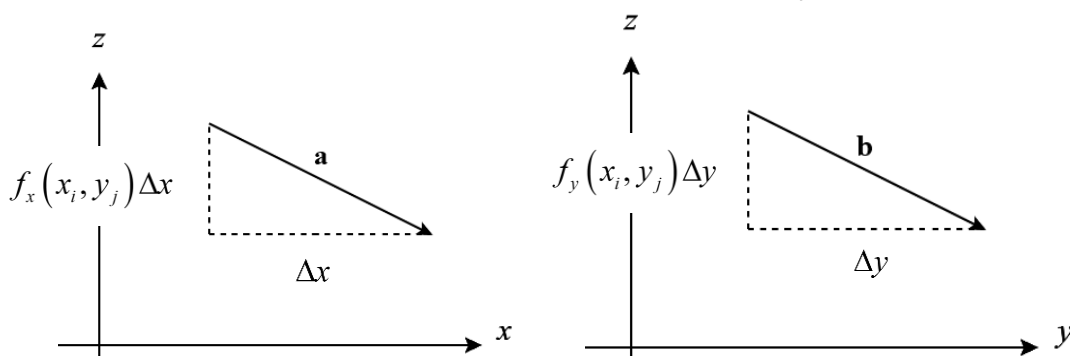
Finding ΔT_{ij} (the area of the portion of the tangent plane we desire) can be done easily using ideas from Chapter 1. Recall that in order to find the area of a parallelogram determined by vectors $\mathbf{a}$ and $\mathbf{b}$ we simply compute $\Delta T_{ij} = \|\mathbf{a} \times \mathbf{b}\|$.

To complete this computation, we need to find the components of $\mathbf{a}$ and $\mathbf{b}$. From the picture we see that

$$\mathbf{a} = \langle \Delta x, 0, z_a \rangle \quad \& \quad \mathbf{b} = \langle 0, \Delta y, z_b \rangle$$



To find the z components we note that both $\mathbf{a}$ and $\mathbf{b}$ share the same direction (3D “slope”) as the tangent lines through the point P_{ij} . Based on our knowledge from Chapter 3, we know these tangent lines have slopes of $f_x(x_i, y_j)$ & $f_y(x_i, y_j)$.



$$\text{and therefore } f_x(x_i, y_j) = \frac{z_a}{\Delta x} \rightarrow z_a = f_x(x_i, y_j) \Delta x \quad \& \quad f_y(x_i, y_j) = \frac{z_b}{\Delta y} \rightarrow z_b = f_y(x_i, y_j) \Delta y$$

$$\text{Therefore, } \mathbf{a} = \langle \Delta x, 0, f_x(x_i, y_j) \Delta x \rangle \quad \& \quad \mathbf{b} = \langle 0, \Delta y, f_y(x_i, y_j) \Delta y \rangle.$$

Now we can compute the area of the parallelogram determined by vectors $\mathbf{a}$ and $\mathbf{b}$.

$$\begin{aligned} \Delta T_{ij} = \|\mathbf{a} \times \mathbf{b}\| &= \left\| \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \Delta x & 0 & f_x(x_i, y_j) \Delta x \\ 0 & \Delta y & f_y(x_i, y_j) \Delta y \end{vmatrix} \right\| = \left\| \langle -f_x(x_i, y_j) \Delta x \Delta y, -f_y(x_i, y_j) \Delta x \Delta y, \Delta x \Delta y \rangle \right\| \\ &= \left\| \langle -f_x(x_i, y_j) \Delta A, -f_y(x_i, y_j) \Delta A, \Delta A \rangle \right\| \\ &= \sqrt{f_x(x_i, y_j)^2 \Delta A^2 + f_y(x_i, y_j)^2 \Delta A^2 + \Delta A^2} \\ &= \left(\sqrt{f_x(x_i, y_j)^2 + f_y(x_i, y_j)^2 + 1} \right) \Delta A. \end{aligned}$$

So, we have that $S = \lim_{m, n \rightarrow \infty} \sum_{i=1}^m \sum_{j=1}^n \Delta T_{ij} = \lim_{m, n \rightarrow \infty} \sum_{i=1}^m \sum_{j=1}^n \left(\sqrt{f_x(x_i, y_j)^2 + f_y(x_i, y_j)^2 + 1} \right) \Delta A$ and the later sum is a double integral!

This, along with a generalization that proves this idea works for surfaces over regions besides rectangles, leads to the following theorem.

Theorem: The area of the surface with equation $z = f(x, y)$ where $(x, y) \in D$ and the first partials of f are continuous is:

$$\begin{aligned} S &= \iint_D \left(\sqrt{(f_x(x, y))^2 + (f_y(x, y))^2 + 1} \right) dA \\ &= \iint_D \left(\sqrt{\left(\frac{\partial z}{\partial x} \right)^2 + \left(\frac{\partial z}{\partial y} \right)^2 + 1} \right) dA. \end{aligned}$$

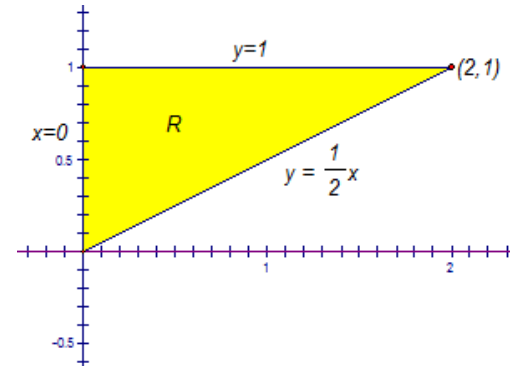
Example 1: Set up but do not evaluate an iterated integral which yields the surface area of the surface determined by $z = 1 + 2x + 3y + 4y^2$ that lies above the triangle with vertices $(0, 0)$, $(1, 1)$ and $(2, 1)$.

A picture of the triangle is shown at the right.

We can represent the region by $D = \left\{ (x, y) \mid 0 \leq x \leq 2, \frac{1}{2}x \leq y \leq 1 \right\}$.

Thus, the surface area will be given by:

$$\iint_D \left(\sqrt{\left(\frac{\partial z}{\partial x} \right)^2 + \left(\frac{\partial z}{\partial y} \right)^2 + 1} \right) dA = \int_0^2 \int_{\frac{1}{2}x}^1 \sqrt{(2)^2 + (8y + 3)^2 + 1} \, dy \, dx.$$



Note: When we are instructed to “set up but do not evaluate”, we stop once we get to an iterated integral that could be plugged into and evaluated by a computer algebra system.

Example 2: Find the exact surface area of the part of the surface given by $z = 4 - x^2 - y^2$ that lies above the xy -plane.

The equation determines an elliptic paraboloid which intersects the xy -plane at the circle

$$0 = 4 - x^2 - y^2 \Rightarrow x^2 + y^2 = 4.$$

Thus, we are interested in finding the surface area of the surface determined by $z = 4 - x^2 - y^2$ which lies above

$$D = \{(x, y) : x^2 + y^2 \leq 4\}.$$

So, the surface area is given by the double integral:

$$\iint_D \left(\sqrt{\left(\frac{\partial z}{\partial x} \right)^2 + \left(\frac{\partial z}{\partial y} \right)^2 + 1} \right) dA = \iint_D \left(\sqrt{(-2x)^2 + (-2y)^2 + 1} \right) dA = \iint_D \left(\sqrt{4x^2 + 4y^2 + 1} \right) dA$$

From the integral we see that it will be favorable if we change to polar coordinates.

We note that in polar $D = \{(r, \theta) \mid 0 \leq \theta \leq 2\pi, 0 \leq r \leq 2\}$. Then,

$$\begin{aligned} \iint_D \left(\sqrt{4x^2 + 4y^2 + 1} \right) dA &= \int_0^{2\pi} \left(\int_0^2 \left(\sqrt{4r^2 + 1} \right) r \, dr \right) d\theta = \int_0^{2\pi} \left(\frac{1}{8} \cdot \frac{2}{3} (4r^2 + 1)^{3/2} \Big|_{r=0}^{r=2} \right) d\theta \\ &= \frac{1}{12} \int_0^{2\pi} (17\sqrt{17} - 1) d\theta = \frac{\pi}{6} (17\sqrt{17} - 1). \end{aligned}$$

Parametric Surfaces and Their Areas

Recall:

In chapter 2 we learned that vector functions of the form $\mathbf{r} : \mathbb{R} \rightarrow \mathbb{R}^3$ is a powerful way to represent curves in the three space. We also learned that vector functions of this form could do a very nice job modeling the movement of a particle in three dimensions.

Then, in chapter 3 we learned that the graph of a function of two variables can be graphed as a surface in three dimensions.

In this section we generalize the ideas presented in chapter 2 to surfaces. This leads to a more powerful way of representing surfaces (than was learned in chapter 3).

Definition: The set of all points in $\mathbb{R}^3$ such that $x = x(u, v)$ $y = y(u, v)$ $z = z(u, v)$ where (u, v) varies throughout some subset, D , of $\mathbb{R}^2$, is called a parametric surface σ and the three equations are called the parametric equations of σ .

Note: Each choice of (u, v) gives a point on σ ; by making all possible choices we get all of σ . A nice illustration of how this works can be found in figure 14.4.5 on page 1028 of the book.

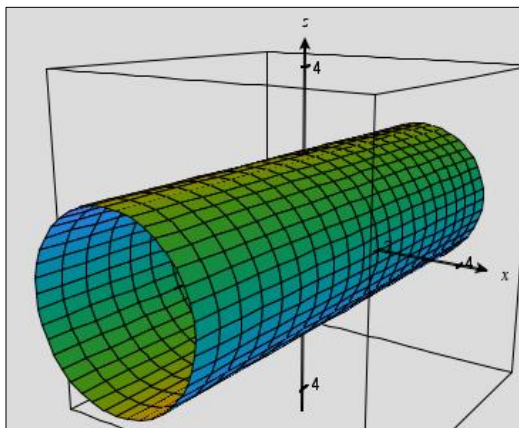
Note: We can represent a parametric surface by a vector function of the form

$\mathbf{r}(u, v) = x(u, v)\mathbf{i} + y(u, v)\mathbf{j} + z(u, v)\mathbf{k}$ where we obtain a picture of the surface by thinking of each output of $\mathbf{r}$ as a position vector and we graph only its terminal point.

Example 3: Identify and sketch the surface given by the vector function

$$\mathbf{r}(u, v) = (2 \cos(u))\mathbf{i} + (v)\mathbf{j} + (2 \sin(u))\mathbf{k}.$$

We note that the parametric equations are $x = 2 \cos(u)$ $y = v$ $z = 2 \sin(u)$. This means that $x^2 + z^2 = 4$ and y can be any real number. Thus, the surface determined by $\mathbf{r}$ is a right circular cylinder with radius 2 which straddles the y -axis. A picture of the surface determined by $\mathbf{r}$ is below.



Note: The instructor will show students how the three dimensional plotter can be used to graph the above surface. Notice in the above picture how the three dimensional plotter uses grid curves (these are curves obtained by holding one of the parameters constant) to depict the surface.

Example 4: Find parametric equations of the plane given by $x + y + z = 1$.

If we let $x = u$ and $y = v$, then $z = 1 - u - v$.

So, parametric equations are $x = u$ $y = v$ $z = 1 - u - v$.

Example 5:

a) What type of quadric surface does $\mathbf{r}(u, v) = \langle u, v, v^2 - u^2 \rangle$ yield?

We note that parametric equations for the surface determined by $\mathbf{r}$ are $x = u$, $y = v$, $z = v^2 - u^2$. So, a single equation whose graph would yield the surface is $z = y^2 - x^2$. From this equation we see that $\mathbf{r}$ yields a hyperbolic paraboloid.

b) Find a vector valued function $\mathbf{r}(u, v)$ which yields the elliptical cylinder given by: $\frac{x^2}{4} + \frac{y^2}{9} = 1$.

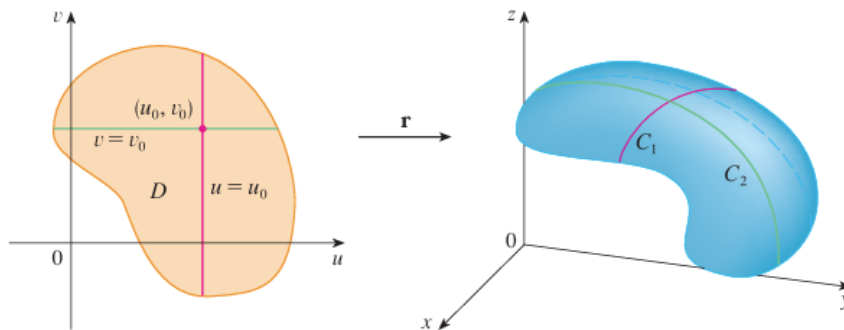
Since the equation provides no restriction on z , we let $z = v$. Then, to satisfy the given equation we let $x = 2\cos(u)$ and $y = 3\sin(u)$. So, $\mathbf{r}(u, v) = \langle 2\cos(u), 3\sin(u), v \rangle$ would yield the elliptical cylinder.

Exploration:

Now, suppose that we would like to find the tangent plane to the surface σ (at some point on σ) where σ is traced out by a vector function $\mathbf{r}(u, v) = x(u, v)\mathbf{i} + y(u, v)\mathbf{j} + z(u, v)\mathbf{k}$.

Specifically suppose we want the tangent plane to the surface at some point P_0 with position vector $\mathbf{r}(u_0, v_0)$.

Suppose we consider the grid curve on σ formed by $\mathbf{r}(u, v)$ when we let $u = u_0$ and the grid curve on σ formed by $\mathbf{r}(u, v)$ when we let $v = v_0$. A picture of the situation is:



The curve on σ in red is the grid curve formed by $\mathbf{r}(u_0, v)$. The curve on σ in green is the grid curve formed by $\mathbf{r}(u, v_0)$. We want the tangent plane to contain both the vectors $\mathbf{r}_v(u_0, v_0)$ and $\mathbf{r}_u(u_0, v_0)$ (since these vectors are tangent to the green and red curves at P_0 , respectively).

Note: $\mathbf{r}_u = \langle x_u, y_u, z_u \rangle$ and $\mathbf{r}_v = \langle x_v, y_v, z_v \rangle$

As long as $\mathbf{r}_u(u_0, v_0) \times \mathbf{r}_v(u_0, v_0) \neq \mathbf{0}$, we can let $\mathbf{r}_u(u_0, v_0) \times \mathbf{r}_v(u_0, v_0)$ be the normal vector to the tangent plane at P_0 .

Definition: Suppose a surface σ is given by $\mathbf{r}(u, v) = x(u, v)\mathbf{i} + y(u, v)\mathbf{j} + z(u, v)\mathbf{k}$. If $\mathbf{r}_u \times \mathbf{r}_v \neq \mathbf{0}$ then the surface σ is called smooth.

Note: Intuitively, a surface is smooth if it has no sharp corners. We can also say that a surface, σ , is smooth if we can find a tangent plane to σ at every point on σ .

Fact: Suppose a surface σ is given by $\mathbf{r}(u, v) = x(u, v)\mathbf{i} + y(u, v)\mathbf{j} + z(u, v)\mathbf{k}$. If $\mathbf{r}(u_0, v_0) = \langle x_0, y_0, z_0 \rangle$ and $\mathbf{r}_u(u_0, v_0) \times \mathbf{r}_v(u_0, v_0) \neq \mathbf{0}$, then an equation of the tangent plane to σ at the point (x_0, y_0, z_0) is

$$(\mathbf{r}_u(u_0, v_0) \times \mathbf{r}_v(u_0, v_0)) \cdot \langle x - x_0, y - y_0, z - z_0 \rangle = 0.$$

Example 6: Find an equation of the tangent plane to the surface given by $\mathbf{r}(u, v) = \langle u^2, v^2, u + 2v \rangle$ at the point $(1, 1, 3)$.

First note $\mathbf{r}(1, 1) = \langle 1, 1, 3 \rangle$. Also, $\mathbf{r}_u = \langle 2u, 0, 1 \rangle$ and $\mathbf{r}_v = \langle 0, 2v, 2 \rangle$.

We then calculate that $\mathbf{r}_u \times \mathbf{r}_v = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 2u & 0 & 1 \\ 0 & 2v & 2 \end{vmatrix} = \langle -2v, -4u, 4uv \rangle$.

This means that $\langle -2(1), -4(1), 4(1)(1) \rangle = \langle -2, -4, 4 \rangle$ is a vector normal to the tangent plane to the surface at $(1, 1, 3)$.

So, an equation for the tangent plane is: $-2(x - 1) - 4(y - 1) + 4(z - 3) = 0 \Leftrightarrow x + 2y - 2z = -3$.

Note: If time permits the instructor will illustrate the above example with a three dimensional plotter.

As we did previously we can approximate the surface area of a surface by using portions of tangent planes, and as our partition gets finer our approximation approaches perfection.

This idea leads to the following definition.

Definition: If a smooth parametric surface σ is given by the equation $\mathbf{r}(u, v) = x(u, v)\mathbf{i} + y(u, v)\mathbf{j} + z(u, v)\mathbf{k}$ where $(u, v) \in D$ and σ is covered just once as (u, v) ranges throughout the parameter domain D , then the surface area of σ is $S = \iint_D \|\mathbf{r}_u \times \mathbf{r}_v\| dA$.

Note: See page 1034-1035 of the book for the Riemann sum version of the above definition.

Example 7: Find the exact surface area of the portion of the spiral ramp given by

$$\mathbf{r}(u, v) = \langle u \cos(v), u \sin(v), v \rangle \quad \text{where } (u, v) \in \{(u, v) : 0 \leq u \leq 1, 0 \leq v \leq u\}$$

If v is constant we note that there are no two u values that will yield the same first and second component.

Thus, we know the surface is covered once for $(u, v) \in \{(u, v) : 0 \leq u \leq 1, 0 \leq v \leq u\}$.

We calculate that:

$$\mathbf{r}_u \times \mathbf{r}_v = \langle \cos(v), \sin(v), 0 \rangle \times \langle -u \sin(v), u \cos(v), 1 \rangle = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \cos(v) & \sin(v) & 0 \\ -u \sin(v) & u \cos(v) & 1 \end{vmatrix} = \langle \sin(v), -\cos(v), u \rangle.$$

So, by the above definition, the surface area is:

$$\begin{aligned} S &= \int_0^1 \int_0^u \|\langle \sin(v), -\cos(v), u \rangle\| dv du = \int_0^1 \int_0^u \sqrt{1+u^2} dv du = \int_0^1 \left(v \sqrt{1+u^2} \Big|_{v=0}^{v=u} \right) du = \int_0^1 u \sqrt{1+u^2} du \\ &= \frac{1}{3} (1+u^2)^{\frac{3}{2}} \Big|_0^1 = \frac{2\sqrt{2}-1}{3}. \end{aligned}$$

Remark: Suppose we want to find the surface area of the surface determined by $z = f(x, y)$ where $(x, y) \in D$.

Rather than using the ideas from the beginning of this section, we can do this with the theory developed about parametric surfaces.

Let $x = u$ & $y = v$ and then $z = f(u, v)$.

So, $\mathbf{r}(u, v) = \langle u, v, f(u, v) \rangle$ where $(u, v) \in E$ (for some region E in the uv -plane) determines the surface we are interested in.

Then, we calculate that

$$\mathbf{r}_u \times \mathbf{r}_v = \langle 1, 0, f_u(u, v) \rangle \times \langle 0, 1, f_v(u, v) \rangle = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 1 & 0 & f_u(u, v) \\ 0 & 1 & f_v(u, v) \end{vmatrix} = \langle -f_u(u, v), -f_v(u, v), 1 \rangle.$$

$$\text{So, } \|\mathbf{r}_u \times \mathbf{r}_v\| = \sqrt{f_u(u, v)^2 + f_v(u, v)^2 + 1}$$

$$\text{Then, by definition the surface area of our surface of interest is } S = \iint_E \sqrt{f_u(u, v)^2 + f_v(u, v)^2 + 1} dA_{uv}$$

But, since $x = u$ & $y = v$ we simply have

$$S = \iint_E \sqrt{f_u(u, v)^2 + f_v(u, v)^2 + 1} dA_{uv} = \iint_D \sqrt{1 + (f_x(x, y))^2 + (f_y(x, y))^2} dA$$

which is the exact formula that we derived in previously for surface area!